

박상훈 (Sanghoon Park)

Busan, South Korea

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Research Interest

Scheduling Manufacturing System via Deep Reinforcement Learning

- Applying deep reinforcement learning algorithms to practical industrial problems

Education

Pusan National University, Busan, South Korea

Mar 2022 – Feb 2027 (Expected)

Bachelor of Business Administration (B.B.A.), Expected Feb 2027

GPA: 2.83 / 4.5 (As of Dec 2025)

Work & Research Experience

Undergraduate Researcher

Dept. of Business Administration (Advisor: Prof. Kyungsu Park)

Oct 2024 – Present

- **[First Author, In Preparation]** *Scheduling Non-Permutation Flow Shop with Missing Operations and Limited Buffers via Asynchronous Branching DRL* (Target Journal: Computers & Industrial Engineering, Expected: Jul 2026)
- Developed a custom Discrete Event Simulation (DES) environment to model complex NPFSP constraints, including Window-based Inspection and limited buffers.
- Proposed A-BD3QN (Asynchronous Branching Dueling Double DQN) architecture to address multi-dimensional action spaces and asynchronous decision-making across machines.
- **Skills:** Python, PyTorch, WandB

Projects

PNU Campus Chatbot | Backend Developer

Nov 2024 – Feb 2025

- Developed an asynchronous API server and database to provide campus information.
- Integrated OpenAI function calling to process natural language queries.
- **Skills:** FastAPI, PostgreSQL, Uvicorn

Review-based Rating Prediction Service | Backend Developer

Sep 2024 – Jan 2025

- Built REST APIs and designed the database for a web service predicting star ratings from text reviews.
- Built an asynchronous task processing API server utilizing Celery and Redis.
- **Skills:** Django REST framework, PostgreSQL, Celery, Redis

Extracurricular Activities & Certification

- **ADsP (Advanced Data Analytics Semi-Professional)** Certification

- **AIDeveloper Club** Mar 2024 – Present
- **Tech Blog (Velog):** Documenting RL algorithms and implementations Oct 2024 – Present

Skills

- **Programming:** Python, C++, C, R
- **Deep Learning & RL:** PyTorch, TensorFlow, WandB
- **Backend & DB:** Django, FastAPI, PostgreSQL, SQLAlchemy, Redis

Awards

- **3rd Place,** Computer Vision Model Implementation Competition (Hosted by AID, PNU)